ZOO 102 Study PDF

100 Objective Exam-Style Questions & Answer Key

A structured exam-preparation companion for ZOO 102, compiling 100 objective questions and answer keys drawn from the full course outline.

Course: Animal Diversity I (ZOO 102)
Department of Zoology, Nnamdi Azikiwe University, Awka (UNIZIK)

Question 1 of 100

Topic: Introduction to Animal Systematics

Q1. Which of the following best defines Systematics?

A. The study of animal behaviour only

B. The scientific study of the diversity of organisms and the evolutionary relationships among them

C. The study of animal habitats alone

D. The classification of plants only

ANSWER KEY

Correct Option: **B** — The scientific study of the diversity of organisms and the evolutionary relationships among them. Systematics involves the identification, naming, description, classification and study of the evolutionary history of organisms.

Question 2 of 100

Topic: Introduction to Animal Systematics

Q2. Taxonomy differs from Systematics in that Taxonomy is mainly concerned with

A. Describing, naming and classifying organisms
B. Studying animal migration patterns
C. Determining the age of fossils
D. Studying animal nutrition

ANSWER KEY

Correct Option: **A** — Describing, naming and classifying organisms. Taxonomy is the theory and practice of describing, naming and classifying organisms, while systematics also covers evolutionary relationships.

Question 3 of 100

Topic: Introduction to Animal Systematics

Q3. Which of the following is NOT among the major goals of Systematics?

A. Discovering and describing biological diversity
B. Inferring phylogenetic relationships among organisms
C. Classifying organisms based on evolutionary relationships
D. Determining the market value of animal products

ANSWER KEY

Correct Option: **D** — Determining the market value of animal products. The three main goals of systematics are discovery/description of diversity, inferring phylogeny, and classification based on relationships.

Question 4 of 100

Topic: Introduction to Animal Systematics

Q4. A named group of organisms of any rank in a classification system is called a

A. Species
B. Taxon
C. Clade only
D. Phenotype

ANSWER KEY

Correct Option: **B** — Taxon. A taxon (plural: taxa) is a named group of organisms at any hierarchical rank, e.g. species, genus, family.

Question 5 of 100

Topic: Introduction to Animal Systematics

Q5. The process of arranging organisms into groups based on shared characteristics and relatedness is known as

A. Nomenclature
B. Classification
C. Identification
D. Cladogenesis

ANSWER KEY

Correct Option: **B** — Classification. Classification is the arrangement of organisms into taxa based on shared characteristics and evolutionary relationships.

Question 6 of 100

Topic: Introduction to Animal Systematics

Q6. Which of the following is an importance of animal systematics?

A. It provides a universal system for naming and identifying organisms
B. It increases the population of endangered species
C. It eliminates the need for conservation
D. It replaces the need for ecological studies

ANSWER KEY

Correct Option: **A** — It provides a universal system for naming and identifying organisms. Systematics provides a universal naming system, establishes evolutionary relationships, and supports biodiversity conservation.

Question 7 of 100

Topic: Introduction to Animal Systematics

Q7. Who is generally regarded as the Father of Taxonomy?

A. Charles Darwin
B. Aristotle
C. Carolus Linnaeus
D. John Ray

ANSWER KEY

Correct Option: **C** — Carolus Linnaeus. Carolus Linnaeus developed the hierarchical classification system and introduced binomial nomenclature.

Question 8 of 100

Topic: History of Animal Diversity

Q8. Aristotle classified about 500 animal species mainly on the basis of

A. DNA sequence similarity
B. Mode of reproduction and habitat
C. Fossil records
D. Geographic distribution only

ANSWER KEY

Correct Option: **B** — Mode of reproduction and habitat. Aristotle grouped animals into Enaima (with blood) and Anaima (without blood) based on reproduction and habitat.

Question 9 of 100

Topic: History of Animal Diversity

Q9. John Ray's major contribution to classification was the introduction of the concept of

A. DNA barcoding
B. Species as a basic unit of classification
C. The molecular clock
D. Cladistics

ANSWER KEY

Correct Option: **B** — Species as a basic unit of classification. John Ray introduced species as a fundamental taxonomic unit and used structural characters for classification.

Question 10 of 100

Topic: History of Animal Diversity

Q10. The hierarchical (Linnaean) system of classification and binomial nomenclature were developed by

A. Charles Darwin
B. Aristotle
C. Carolus Linnaeus
D. Alfred Wallace

ANSWER KEY

Correct Option: **C** — Carolus Linnaeus. Linnaeus published his hierarchical classification and binomial system in Systema Naturae (1758).

Question 11 of 100

Topic: History of Animal Diversity

Q11. Charles Darwin influenced modern systematics chiefly through his

A. Theory of evolution by natural selection
B. Invention of the microscope
C. Discovery of DNA
D. Development of the ICZN

ANSWER KEY

Correct Option: **A** — Theory of evolution by natural selection. Darwin's theory of evolution provided the biological basis for classifying organisms by evolutionary relatedness.

Question 12 of 100

Topic: History of Animal Diversity

Q12. In which publication did Linnaeus first consistently apply binomial names to animals?

A. On the Origin of Species
B. Systema Naturae, 10th edition (1758)
C. Historia Animalium
D. Philosophiae Naturalis

ANSWER KEY

Correct Option: **B** — Systema Naturae, 10th edition (1758). The 10th edition of Systema Naturae (1758) is the accepted starting point of zoological nomenclature.

Question 13 of 100

Topic: History of Animal Diversity

Q13. Aristotle is generally regarded as the

A. Father of Genetics
B. Father of Zoology
C. Father of Ecology
D. Father of Cladistics

ANSWER KEY

Correct Option: **B** — Father of Zoology. Aristotle is considered the father of Zoology for his early systematic study and classification of animals.

Question 14 of 100

Topic: Basis of Categorisation of Animal Diversity (Symmetry)

Q14. Symmetry in animal classification refers to

A. The colour pattern of an organism
B. The balanced arrangement of body parts around a central axis or plane
C. The number of legs an animal has
D. The type of habitat an animal lives in

ANSWER KEY

Correct Option: **B** — The balanced arrangement of body parts around a central axis or plane. Symmetry describes how body parts are arranged so the body can be divided into corresponding parts.

Question 15 of 100

Topic: Basis of Categorisation of Animal Diversity (Symmetry)

Q15. Which type of symmetry is exhibited by Hydra and sea anemones?

A. Bilateral symmetry
B. Asymmetry
C. Radial symmetry
D. Spherical symmetry

ANSWER KEY

Correct Option: **C** — Radial symmetry. Radial symmetry allows any plane through the central axis to divide the body into similar halves, as in Hydra.

Question 16 of 100

Topic: Basis of Categorisation of Animal Diversity (Symmetry)

Q16. An animal that can only be divided into mirror-image left and right halves by a single plane exhibits

A. Radial symmetry
B. Bilateral symmetry
C. Biradial symmetry
D. Asymmetry

ANSWER KEY

Correct Option: **B** — Bilateral symmetry. Bilateral symmetry, seen in earthworms and humans, allows division into mirror-image halves along one plane only.

Question 17 of 100

Topic: Basis of Categorisation of Animal Diversity (Symmetry)

Q17. Most members of Phylum Porifera (sponges) exhibit which form of symmetry?

A. Bilateral symmetry
B. Radial symmetry
C. Asymmetry
D. Biradial symmetry

ANSWER KEY

Correct Option: **C** — Asymmetry. Most sponges lack any definite plane or axis of symmetry and are therefore described as asymmetrical.

Question 18 of 100

Topic: Basis of Categorisation of Animal Diversity (Symmetry)

Q18. Spherical symmetry, in which any plane through the centre divides the body into equal halves, is found in

A. Earthworms
B. Some Protozoa such as Heliozoans
C. Sponges
D. Insects

ANSWER KEY

Correct Option: **B** — Some Protozoa such as Heliozoans. Spherical symmetry occurs in some protozoans such as Heliozoans and radiolarians.

Question 19 of 100

Topic: Basis of Categorisation of Animal Diversity (Symmetry)

Q19. Bilateral symmetry is closely associated with which evolutionary trend?

A. Cephalisation and directional movement
B. Loss of nervous tissue
C. Sessile lifestyle only
D. Reduction of sensory organs

ANSWER KEY

Correct Option: **A** — Cephalisation and directional movement. Bilateral symmetry favours cephalisation (concentration of sense organs anteriorly) and active, directional movement.

Question 20 of 100

Topic: Basis of Categorisation of Animal Diversity (Symmetry)

Q20. Biradial symmetry, a modification of radial symmetry in which only two planes divide the body into mirror-image halves, is seen in

A. Ctenophores (comb jellies)
B. Earthworms
C. Sponges
D. Insects

ANSWER KEY

Correct Option: **A** — Ctenophores (comb jellies). Biradial symmetry is characteristic of ctenophores, where only two specific planes produce mirror-image halves.

Question 21 of 100

Topic: General Classification of the Animal Kingdom

Q21. Which of the following represents the correct order of taxonomic ranks from highest to lowest?

A. Species, Genus, Family, Order, Class, Phylum, Kingdom
B. Kingdom, Phylum, Class, Order, Family, Genus, Species
C. Phylum, Kingdom, Class, Family, Order, Genus, Species
D. Kingdom, Class, Phylum, Order, Family, Genus, Species

ANSWER KEY

Correct Option: **B** — Kingdom, Phylum, Class, Order, Family, Genus, Species. The standard hierarchy runs from Kingdom (highest) down to Species (lowest): Kingdom-Phylum-Class-Order-Family-Genus-Species.

Question 22 of 100

Topic: General Classification of the Animal Kingdom

Q22. Animals that possess a vertebral column (backbone) are classified as

A. Invertebrates
B. Vertebrates
C. Protostomes only
D. Diploblastic animals

ANSWER KEY

Correct Option: **B** — Vertebrates. Vertebrates possess a vertebral column and internal skeleton, distinguishing them from invertebrates.

Question 23 of 100

Topic: General Classification of the Animal Kingdom

Q23. In Protostomes, the blastopore develops into the

A. Anus
B. Mouth
C. Coelom
D. Notochord

ANSWER KEY

Correct Option: **B** — Mouth. In protostomes (e.g. Annelida, Mollusca, Arthropoda), the blastopore becomes the mouth of the adult.

Question 24 of 100

Topic: General Classification of the Animal Kingdom

Q24. In Deuterostomes, the blastopore develops into the

A. Mouth
B. Anus
C. Nerve cord
D. Coelom lining

ANSWER KEY

Correct Option: **B** — Anus. In deuterostomes (e.g. Echinodermata, Chordata), the blastopore becomes the anus, while the mouth forms secondarily.

Question 25 of 100

Topic: General Classification of the Animal Kingdom

Q25. Animals possessing only two germ layers (ectoderm and endoderm) are described as

A. Triploblastic
B. Diploblastic
C. Coelomate
D. Pseudocoelomate

ANSWER KEY

Correct Option: **B** — Diploblastic. Diploblastic animals, such as Cnidarians, have two germ layers with a non-cellular mesoglea between them.

Question 26 of 100

Topic: General Classification of the Animal Kingdom

Q26. Animals with a body cavity that is not completely lined by mesoderm are called

A. Acoelomates
B. Coelomates
C. Pseudocoelomates
D. Diploblastic animals

ANSWER KEY

Correct Option: **C** — Pseudocoelomates. Pseudocoelomates, such as Nematoda, have a body cavity that is only partly lined by mesoderm.

Question 27 of 100

Topic: General Classification of the Animal Kingdom

Q27. Which of the following criteria is NOT typically used in the general classification of the animal kingdom?

A. Body symmetry
B. Number of germ layers
C. Presence or absence of a body cavity
D. Colour of the animal

ANSWER KEY

Correct Option: **D** — Colour of the animal. Classification is based on symmetry, germ layers, body cavity type, segmentation and development, not colour.

Question 28 of 100

Topic: Binomial Nomenclature

Q28. Binomial Nomenclature refers to the system of naming organisms using

A. A single common name
B. Two Latinised names - genus and species
C. Three names for every organism
D. Only the family name

ANSWER KEY

Correct Option: **B** — Two Latinised names - genus and species. Binomial nomenclature uses two names: the genus (capitalised) and the specific epithet (lowercase), e.g. Homo sapiens.

Question 29 of 100

Topic: Binomial Nomenclature

Q29. The binomial system of nomenclature was introduced by

A. Charles Darwin
B. Carolus Linnaeus
C. Aristotle
D. John Ray

ANSWER KEY

Correct Option: **B** — Carolus Linnaeus. Linnaeus introduced binomial nomenclature in Systema Naturae, 10th edition (1758).

Question 30 of 100

Topic: Binomial Nomenclature

Q30. In writing a scientific name, which rule is correct?

A. Both genus and species names begin with capital letters

B. The genus name begins with a capital letter and species epithet with a lowercase letter

C. The species epithet begins with a capital letter and genus with lowercase

D. Scientific names are never italicised

ANSWER KEY

Correct Option: **B** — The genus name begins with a capital letter and species epithet with a lowercase letter. Correct format capitalises only the genus, keeps the species epithet lowercase, and italicises both, e.g. Panthera leo.

Question 31 of 100

Topic: Binomial Nomenclature

Q31. The main advantage of binomial nomenclature is that it

A. Varies from country to country
B. Provides a universal, standardised name for each species
C. Uses only common names
D. Is only applicable to plants

ANSWER KEY

Correct Option: **B** — Provides a universal, standardised name for each species. Binomial names are universally recognised, avoiding confusion caused by differing common names across regions.

Question 32 of 100

Topic: Binomial Nomenclature

Q32. Trinomial nomenclature, such as Panthera leo leo, is used to designate

A. A genus
B. A family
C. A subspecies
D. A phylum

ANSWER KEY

Correct Option: **C** — A subspecies. Trinomial nomenclature adds a third name to denote a subspecies within a species.

Question 33 of 100

Topic: Binomial Nomenclature

Q33. Which of the following is an example of a correctly written scientific name?

A. homo Sapiens
B. Homo sapiens
C. HOMO SAPIENS
D. Sapiens homo

ANSWER KEY

Correct Option: **B** — Homo sapiens. The correct format capitalises the genus only, keeps species lowercase, and lists genus before species: Homo sapiens.

Question 34 of 100

Topic: International Rules of Zoological Nomenclature (Brief Account)

Q34. The acronym ICZN stands for

A. International Code for Zoo Naming
B. International Code of Zoological Nomenclature
C. Institute for Classification of Zoological Names
D. International Committee for Zoology and Nature

ANSWER KEY

Correct Option: **B** — International Code of Zoological Nomenclature. ICZN stands for the International Code of Zoological Nomenclature, which governs formal animal naming.

Question 35 of 100

Topic: International Rules of Zoological Nomenclature (Brief Account)

Q35. The primary objective of the ICZN is to

A. Promote stability and universality of scientific animal names
B. Determine animal habitats
C. Regulate animal breeding
D. Classify plants

ANSWER KEY

Correct Option: **A** — Promote stability and universality of scientific animal names. The ICZN ensures that each animal name is unique, unambiguous, and stable across the scientific community.

Question 36 of 100

Topic: International Rules of Zoological Nomenclature (Brief Account)

Q36. According to the Principle of Priority, the valid name of a taxon is

A. The most recently published name
B. The most popular name
C. The earliest validly published name
D. The name chosen by majority vote

ANSWER KEY

Correct Option: **C** — The earliest validly published name. The Principle of Priority gives precedence to the earliest validly published name under the Code.

Question 37 of 100

Topic: International Rules of Zoological Nomenclature (Brief Account)

Q37. Two or more different scientific names applied to the same taxon are called

A. Homonyms
B. Synonyms
C. Type species
D. Cladograms

ANSWER KEY

Correct Option: **B** — Synonyms. Synonyms are different names for the same taxon; the earliest is the valid (senior) synonym.

Question 38 of 100

Topic: International Rules of Zoological Nomenclature (Brief Account)

Q38. A name that is spelled identically to another name but applied to a different taxon is called a

A. Synonym
B. Homonym
C. Trinomial
D. Plesiomorphy

ANSWER KEY

Correct Option: **B** — Homonym. A homonym is an identical name used for different taxa; only the oldest (senior homonym) remains valid.

Question 39 of 100

Topic: International Rules of Zoological Nomenclature (Brief Account)

Q39. According to the Principle of Binomial Nomenclature under the ICZN, a species name must consist of

A. One word only
B. Two names - genus and species
C. Three or more names
D. A number and a letter

ANSWER KEY

Correct Option: **B** — Two names - genus and species. The Code stipulates that species names are combinations of exactly two names - generic and specific.

Question 40 of 100

Topic: International Rules of Zoological Nomenclature (Brief Account)

Q40. Which of the following is one of the basic principles of the ICZN?

A. Principle of Priority
B. Principle of Photosynthesis
C. Principle of Natural Selection
D. Principle of Symbiosis

ANSWER KEY

Correct Option: **A** — Principle of Priority. The ICZN is founded on principles including Priority, Binomial Nomenclature, Coordination and Homonymy.

Question 41 of 100

Topic: New Trends in Systematics: Numerical Taxonomy and Cladistics

Q41. Numerical Taxonomy (Phenetics) classifies organisms based mainly on

A. Their evolutionary ancestry only
B. Overall similarity of observable characteristics
C. DNA sequence alone
D. Their geographical location

ANSWER KEY

Correct Option: **B** — Overall similarity of observable characteristics. Phenetics groups organisms using overall phenotypic similarity, analysed statistically, regardless of evolutionary history.

Question 42 of 100

Topic: New Trends in Systematics: Numerical Taxonomy and Cladistics

Q42. Cladistics classifies organisms strictly according to

A. Overall phenotypic similarity
B. Shared derived characters indicating common ancestry
C. Habitat type
D. Body size

ANSWER KEY

Correct Option: **B** — Shared derived characters indicating common ancestry. Cladistics groups organisms by shared derived characteristics (synapomorphies) reflecting recent common ancestry.

Question 43 of 100

Topic: New Trends in Systematics: Numerical Taxonomy and Cladistics

Q43. A branching diagram used in cladistics to show hypothesised evolutionary relationships is called a

A. Phenogram
B. Cladogram
C. Dendrogram of similarity
D. Taxonomic key

ANSWER KEY

Correct Option: **B** — Cladogram. A cladogram is the tree-like diagram used to represent relationships based on shared derived characters.

Question 44 of 100

Topic: New Trends in Systematics: Numerical Taxonomy and Cladistics

Q44. Structures that are similar because of shared evolutionary ancestry, such as vertebrate forelimbs, are described as

A. Analogous
B. Homologous
C. Convergent
D. Polyphyletic

ANSWER KEY

Correct Option: **B** — Homologous. Homologous structures share a common evolutionary origin, unlike analogous structures which arise by convergence.

Question 45 of 100

Topic: New Trends in Systematics: Numerical Taxonomy and Cladistics

Q45. Wings in insects and birds, which perform a similar function but evolved independently, are examples of

A. Homologous structures
B. Analogous structures
C. Synapomorphies
D. Plesiomorphies

ANSWER KEY

Correct Option: **B** — Analogous structures. Analogous structures perform similar functions but do not share a common evolutionary origin (convergent evolution).

Question 46 of 100

Topic: New Trends in Systematics: Numerical Taxonomy and Cladistics

Q46. A shared derived character state used to define a clade is known as a

A. Plesiomorphy
B. Synapomorphy
C. Homoplasy only
D. Phenetic index

ANSWER KEY

Correct Option: **B** — Synapomorphy. A synapomorphy is a derived character shared by two or more taxa and their common ancestor, used to define clades.

Question 47 of 100

Topic: New Trends in Systematics: Numerical Taxonomy and Cladistics

Q47. A group that includes taxa from different ancestral lineages that do not share a recent common ancestor is called

A. Monophyletic
B. Paraphyletic
C. Polyphyletic
D. Homologous

ANSWER KEY

Correct Option: **C** — Polyphyletic. A polyphyletic group includes distantly related taxa, usually grouped due to convergent (analogous) traits.

Question 48 of 100

Topic: Molecular Systematics

Q48. Molecular Systematics uses information from which of the following to study evolutionary relationships?

A. Fossil records only
B. DNA, RNA and protein sequences
C. Animal behaviour only
D. Habitat type only

ANSWER KEY

Correct Option: **B** — DNA, RNA and protein sequences. Molecular systematics compares nucleotide and amino acid sequences to infer evolutionary relationships.

Question 49 of 100

Topic: Molecular Systematics

Q49. An advantage of molecular systematics over purely morphological classification is that molecular data

A. Are always identical across all species
B. Provide many independent characters less affected by convergent evolution
C. Cannot be used for distantly related organisms
D. Require no laboratory analysis

ANSWER KEY

Correct Option: **B** — Provide many independent characters less affected by convergent evolution. Molecular sequences provide numerous quantifiable characters that are less prone to convergence than morphology.

Question 50 of 100

Topic: Molecular Systematics

Q50. The process of determining the exact order of nucleotide bases in a DNA molecule is called

A. DNA replication
B. DNA sequencing
C. DNA transcription
D. DNA barcoding only

ANSWER KEY

Correct Option: **B** — DNA sequencing. DNA sequencing determines nucleotide order, and the sequences are compared to estimate relatedness among taxa.

Question 51 of 100

Topic: Molecular Systematics

Q51. The hypothesis that mutations accumulate in DNA/protein at a relatively constant rate over time is known as the

A. Founder effect
B. Molecular Clock hypothesis
C. Hardy-Weinberg principle
D. Bottleneck effect

ANSWER KEY

Correct Option: **B** — Molecular Clock hypothesis. The Molecular Clock hypothesis allows estimation of divergence times by comparing accumulated sequence differences.

Question 52 of 100

Topic: Molecular Systematics

Q52. Which gene is commonly used as the standard marker for DNA barcoding in animals?

A. 18S rRNA
B. Mitochondrial cytochrome c oxidase I (COI)
C. Ribulose bisphosphate carboxylase
D. Beta-globin gene

ANSWER KEY

Correct Option: **B** — Mitochondrial cytochrome c oxidase I (COI). The mitochondrial COI gene is the standard barcode region used for animal species identification.

Question 53 of 100

Topic: Molecular Systematics

Q53. DNA Barcoding is best described as a technique used to

A. Identify species using a short standardised gene sequence
B. Determine an animal's age
C. Measure an animal's body temperature
D. Classify plants only

ANSWER KEY

Correct Option: **A** — Identify species using a short standardised gene sequence. DNA barcoding matches a short standardised sequence from an unknown specimen against a reference library to identify species.

Question 54 of 100

Topic: Major Features of an Animal Phylogenetic Tree

Q54. A phylogenetic tree is best described as a diagram that represents

A. The geographic range of a species
B. The hypothesised evolutionary relationships and history of descent among organisms
C. The habitat preference of an organism
D. The digestive system of an animal

ANSWER KEY

Correct Option: **B** — The hypothesised evolutionary relationships and history of descent among organisms. A phylogenetic tree is a branching diagram showing evolutionary relationships based on shared characteristics.

Question 55 of 100

Topic: Major Features of an Animal Phylogenetic Tree

Q55. In a phylogenetic tree, the point representing the common ancestor of all included taxa is called the

A. Node
B. Root
C. Tip
D. Branch

ANSWER KEY

Correct Option: **B** — Root. The root represents the common ancestor of all taxa in the tree, from which all branches originate.

Question 56 of 100

Topic: Major Features of an Animal Phylogenetic Tree

Q56. A point of divergence where a lineage splits into two or more descendant lineages is called a

A. Root
B. Node
C. Outgroup
D. Tip

ANSWER KEY

Correct Option: **B** — Node. A node marks a hypothetical common ancestor where a lineage splits into descendant branches.

Question 57 of 100

Topic: Major Features of an Animal Phylogenetic Tree

Q57. In a phylogenetic tree, a present-day taxon located at the end of a branch is referred to as a

A. Root
B. Node
C. Tip (leaf)
D. Outgroup

ANSWER KEY

Correct Option: **C** — Tip (leaf). A tip (or leaf) represents a present-day taxon at the terminal end of a branch.

Question 58 of 100

Topic: Major Features of an Animal Phylogenetic Tree

Q58. A taxon used as a point of reference to root a phylogenetic tree and determine character polarity is called the

A. Ingroup
B. Outgroup
C. Sister group only
D. Clade

ANSWER KEY

Correct Option: **B** — Outgroup. The outgroup is a less closely related taxon used to root the tree and determine direction of character change.

Question 59 of 100

Topic: Major Features of an Animal Phylogenetic Tree

Q59. The main difference between a rooted and an unrooted phylogenetic tree is that a rooted tree

A. Has no branches
B. Shows a designated common ancestor and direction of evolutionary time
C. Cannot include an outgroup
D. Only applies to plants

ANSWER KEY

Correct Option: **B** — Shows a designated common ancestor and direction of evolutionary time. A rooted tree designates a single common ancestor and shows the direction of evolution, unlike an unrooted tree.

Question 60 of 100

Topic: Diversity of Animal Forms, Structures and Functions

Q60. Animal diversity refers to the variety of animal species differing in

A. Body forms, structures, physiological functions and habitats
B. Only their common names
C. Only their geographic origin
D. Only their size

ANSWER KEY

Correct Option: **A** — Body forms, structures, physiological functions and habitats. Animal diversity encompasses variation in body form, structure, physiology, habitat and behaviour.

Question 61 of 100

Topic: Diversity of Animal Forms, Structures and Functions

Q61. The streamlined body and fins of fish are an example of

A. Convergent behaviour
B. Structural adaptation for efficient movement in water
C. Random mutation with no benefit
D. Analogous digestion

ANSWER KEY

Correct Option: **B** — Structural adaptation for efficient movement in water. Fish body form is structurally adapted for efficient locomotion in an aquatic environment.

Question 62 of 100

Topic: Diversity of Animal Forms, Structures and Functions

Q62. Which of the following is NOT a major factor responsible for animal diversity?

A. Adaptation to different habitats
B. Natural selection and evolution
C. Genetic variation/mutation
D. Uniform environmental conditions worldwide

ANSWER KEY

Correct Option: **D** — Uniform environmental conditions worldwide. Animal diversity arises from adaptation, natural selection, genetic variation and diverse ecological niches, not uniform conditions.

Question 63 of 100

Topic: Diversity of Animal Forms, Structures and Functions

Q63. A heritable characteristic that improves an organism's ability to survive and reproduce in its environment is called

A. Mutation
B. Adaptation
C. Convergence
D. Symmetry

ANSWER KEY

Correct Option: **B** — Adaptation. Adaptation is any heritable trait that enhances survival and reproductive success in a given environment.

Question 64 of 100

Topic: Diversity of Animal Forms, Structures and Functions

Q64. When unrelated organisms independently evolve similar structures due to similar environmental pressures, this is called

A. Divergent evolution
B. Convergent evolution
C. Cladogenesis
D. Homology

ANSWER KEY

Correct Option: **B** — Convergent evolution. Convergent evolution occurs when unrelated species develop similar traits, e.g. wings in birds and bats.

Question 65 of 100

Topic: Diversity of Animal Forms, Structures and Functions

Q65. The basic structural design or organisation of an animal's body, generally consistent within a phylum, is called its

A. Genotype
B. Body Plan (Bauplan)
C. Phenotype ratio
D. Karyotype

ANSWER KEY

Correct Option: **B** — Body Plan (Bauplan). The Bauplan (body plan) refers to the fundamental structural organisation of a phylum, including symmetry and body cavity type.

Question 66 of 100

Topic: Diversity of Animal Forms, Structures and Functions

Q66. Wings, as a structural adaptation for flight, are best associated with which mode of life?

A. Aquatic
B. Aerial
C. Burrowing
D. Sessile

ANSWER KEY

Correct Option: **B** — Aerial. Wings are structural adaptations that enable flight in aerial animals such as birds and insects.

Question 67 of 100

Topic: Multicellularity and Development of Embryonic Layers

Q67. An organism composed of a single cell that performs all life processes, such as Amoeba, is described as

A. Multicellular
B. Unicellular
C. Diploblastic
D. Coelomate

ANSWER KEY

Correct Option: **B** — Unicellular. Unicellular organisms consist of a single cell carrying out all life functions, unlike multicellular organisms.

Question 68 of 100

Topic: Multicellularity and Development of Embryonic Layers

Q68. One major advantage of multicellularity is that it allows for

A. Loss of specialisation
B. Cell specialisation and division of labour
C. Reduced body size
D. Elimination of reproduction

ANSWER KEY

Correct Option: **B** — Cell specialisation and division of labour. Multicellularity permits cells to specialise for specific functions, increasing overall efficiency.

Question 69 of 100

Topic: Multicellularity and Development of Embryonic Layers

Q69. A germ layer is best defined as

A. A layer of skin cells only
B. A primary layer of cells formed during gastrulation from which body tissues develop
C. A protective covering of an egg
D. A type of muscle tissue

ANSWER KEY

Correct Option: **B** — A primary layer of cells formed during gastrulation from which body tissues develop. Germ layers form during gastrulation and give rise to all tissues and organs of the adult body.

Question 70 of 100

Topic: Multicellularity and Development of Embryonic Layers

Q70. The skin epidermis and nervous system of an animal develop from the

A. Endoderm
B. Mesoderm
C. Ectoderm
D. Blastocoel

ANSWER KEY

Correct Option: **C** — Ectoderm. The ectoderm gives rise to the epidermis and nervous system in developing embryos.

Question 71 of 100

Topic: Multicellularity and Development of Embryonic Layers

Q71. Muscles and the skeletal system of an animal are derived from the

A. Ectoderm
B. Mesoderm
C. Endoderm
D. Blastopore

ANSWER KEY

Correct Option: **B** — Mesoderm. The mesoderm gives rise to muscles, the skeletal system, circulatory system and gonads.

Question 72 of 100

Topic: Multicellularity and Development of Embryonic Layers

Q72. The embryonic process by which the blastula reorganises to form the germ layers is called

A. Cleavage
B. Gastrulation
C. Fertilisation
D. Ovulation

ANSWER KEY

Correct Option: **B** — Gastrulation. Gastrulation is the process that produces the germ layers (ectoderm, mesoderm, endoderm) from the blastula.

Question 73 of 100

Topic: Multicellularity and Development of Embryonic Layers

Q73. Animals such as Cnidarians (e.g. Hydra), which possess only ectoderm and endoderm, are described as

A. Triploblastic
B. Diploblastic
C. Coelomate
D. Pseudocoelomate

ANSWER KEY

Correct Option: **B** — Diploblastic. Diploblastic animals have two germ layers (ectoderm and endoderm) separated by a non-cellular mesoglea.

Question 74 of 100

Topic: Developmental Modes and the Fate of the Blastopore

Q74. The blastopore is best defined as

A. The outer covering of the egg
B. The opening formed during gastrulation connecting the archenteron to the exterior
C. A type of larval stage
D. The site of fertilisation

ANSWER KEY

Correct Option: **B** — The opening formed during gastrulation connecting the archenteron to the exterior. The blastopore is the opening formed during gastrulation linking the primitive gut (archenteron) to the exterior.

Question 75 of 100

Topic: Developmental Modes and the Fate of the Blastopore

Q75. In protostome development, the blastopore becomes the

A. Anus
B. Mouth
C. Notochord
D. Coelom

ANSWER KEY

Correct Option: **B** — Mouth. In protostomes such as Annelida and Mollusca, the blastopore develops into the adult mouth.

Question 76 of 100

Topic: Developmental Modes and the Fate of the Blastopore

Q76. In deuterostome development, the blastopore becomes the

A. Mouth
B. Anus
C. Gill slit
D. Nerve cord

ANSWER KEY

Correct Option: **B** — Anus. In deuterostomes such as Echinodermata and Chordata, the blastopore develops into the anus, with the mouth forming secondarily.

Question 77 of 100

Topic: Developmental Modes and the Fate of the Blastopore

Q77. Determinate (mosaic) cleavage, in which each embryonic cell's fate is fixed very early, is typical of

A. Deuterostomes
B. Protostomes
C. Diploblastic animals only
D. Pseudocoelomates only

ANSWER KEY

Correct Option: **B** — Protostomes. Protostomes exhibit determinate/mosaic cleavage, unlike deuterostomes which show indeterminate/regulative cleavage.

Question 78 of 100

Topic: Developmental Modes and the Fate of the Blastopore

Q78. In Protostomes, the coelom typically forms by a process known as

A. Enterocoely
B. Schizocoely
C. Gastrulation only
D. Cleavage

ANSWER KEY

Correct Option: **B** — Schizocoely. In protostomes, the coelom forms by schizocoely, where the mesoderm splits to create the coelomic cavity.

Question 79 of 100

Topic: Developmental Modes and the Fate of the Blastopore

Q79. In Deuterostomes, the coelom typically forms by a process known as

A. Schizocoely
B. Enterocoely
C. Blastulation
D. Neurulation

ANSWER KEY

Correct Option: **B** — Enterocoely. In deuterostomes, the coelom forms by enterocoely, where mesodermal pouches pinch off from the archenteron wall.

Question 80 of 100

Topic: Developmental Modes and the Fate of the Blastopore

Q80. Which set of features correctly describes Deuterostome development?

A. Spiral cleavage, blastopore becomes mouth, schizocoely
B. Radial cleavage, blastopore becomes anus, enterocoely
C. Determinate cleavage, blastopore becomes mouth, enterocoely
D. Radial cleavage, blastopore becomes mouth, schizocoely

ANSWER KEY

Correct Option: **B** — Radial cleavage, blastopore becomes anus, enterocoely. Deuterostomes show radial, indeterminate cleavage, the blastopore becomes the anus, and the coelom forms by enterocoely.

Question 81 of 100

Topic: Organisation of Tissues and the Body Cavity

Q81. A tissue is best defined as

A. A single isolated cell
B. A group of similar cells working together to perform a specific function
C. An entire organism
D. A type of body cavity

ANSWER KEY

Correct Option: **B** — A group of similar cells working together to perform a specific function. A tissue consists of similar cells and their intercellular substance working together for a common function.

Question 82 of 100

Topic: Organisation of Tissues and the Body Cavity

Q82. Which of the following correctly represents the levels of organisation in the animal body, from simplest to most complex?

A. Organ, Tissue, Cell, Organ system
B. Cell, Tissue, Organ, Organ system
C. Tissue, Cell, Organ system, Organ
D. Organ system, Organ, Tissue, Cell

ANSWER KEY

Correct Option: **B** — Cell, Tissue, Organ, Organ system. The correct sequence is Cell to Tissue to Organ to Organ system, representing increasing complexity.

Question 83 of 100

Topic: Organisation of Tissues and the Body Cavity

Q83. A fluid-filled body cavity completely lined by mesoderm-derived tissue is called a

A. Pseudocoelom
B. Coelom
C. Blastocoel
D. Archenteron

ANSWER KEY

Correct Option: **B** — Coelom. A true coelom is a body cavity lying between the body wall and gut, fully lined by mesoderm.

Question 84 of 100

Topic: Organisation of Tissues and the Body Cavity

Q84. Flatworms (Platyhelminthes), which lack any body cavity between the body wall and gut, are classified as

A. Coelomates
B. Pseudocoelomates
C. Acoelomates
D. Diploblastic animals

ANSWER KEY

Correct Option: **C** — Acoelomates. Acoelomates such as Platyhelminthes have no body cavity; the space is filled with mesenchyme.

Question 85 of 100

Topic: Organisation of Tissues and the Body Cavity

Q85. Which of the following is NOT a function of the coelom?

A. Providing space for development and movement of internal organs
B. Acting as a hydrostatic skeleton
C. Allowing circulation of materials/wastes
D. Producing chlorophyll for photosynthesis

ANSWER KEY

Correct Option: **D** — Producing chlorophyll for photosynthesis. The coelom aids organ movement, acts as a hydrostatic skeleton, and allows circulation; it plays no role in photosynthesis.

Question 86 of 100

Topic: Organisation of Tissues and the Body Cavity

Q86. The four basic types of animal tissue are

A. Epithelial, Connective, Muscular and Nervous
B. Vascular, Ground, Dermal and Meristematic
C. Epidermal, Cortical, Vascular and Pith
D. Bone, Blood, Fat and Skin

ANSWER KEY

Correct Option: **A** — Epithelial, Connective, Muscular and Nervous. Animal tissues are classified into Epithelial, Connective, Muscular and Nervous tissue.

Question 87 of 100

Topic: Organisation of Tissues and the Body Cavity

Q87. A structure composed of two or more different tissues working together to perform a specific function, such as the heart, is called an

A. Organ system
B. Organ
C. Organelle
D. Epithelium

ANSWER KEY

Correct Option: **B** — Organ. An organ is made up of two or more tissue types working together, e.g. the heart; several organs form an organ system.

Question 88 of 100

Topic: Geographical Distribution of Animal Life

Q88. The branch of biology that studies the geographical distribution of animal species is called

A. Ecology
B. Zoogeography
C. Physiology
D. Taxonomy

ANSWER KEY

Correct Option: **B** — Zoogeography. Zoogeography studies the distribution patterns of animals and the factors that determine them.

Question 89 of 100

Topic: Geographical Distribution of Animal Life

Q89. The system of zoogeographical realms was largely developed by

A. Carolus Linnaeus
B. Philip Lutley Sclater and Alfred Russel Wallace
C. Charles Darwin alone
D. Aristotle

ANSWER KEY

Correct Option: **B** — Philip Lutley Sclater and Alfred Russel Wallace. Sclater proposed the original scheme, later refined by Wallace, giving six major zoogeographical realms.

Question 90 of 100

Topic: Geographical Distribution of Animal Life

Q90. How many major zoogeographical realms are generally recognised worldwide?

A. Four
B. Five
C. Six
D. Eight

ANSWER KEY

Correct Option: **C** — Six. The six major realms are Palaearctic, Nearctic, Neotropical, Ethiopian (Afrotropical), Oriental and Australian.

Question 91 of 100

Topic: Geographical Distribution of Animal Life

Q91. Nigeria belongs to which zoogeographical realm?

- A. Palaearctic
- B. Nearctic
- C. Ethiopian (Afrotropical)
- D. Australian

ANSWER KEY

Correct Option: **C** — Ethiopian (Afrotropical). Nigeria falls within the Ethiopian (Afrotropical) realm, home to animals such as the African elephant and lion.

Question 92 of 100

Topic: Geographical Distribution of Animal Life

Q92. Which of the following is NOT a major factor influencing the geographical distribution of animals?

A. Climate
B. Physical/geographical barriers
C. Availability of food and habitat
D. Animal's favourite colour

ANSWER KEY

Correct Option: **D** — Animal's favourite colour. Distribution is influenced by climate, barriers, food/habitat availability and historical factors such as continental drift.

Question 93 of 100

Topic: Geographical Distribution of Animal Life

Q93. A species that is naturally found only in a particular restricted geographical area is described as

A. Cosmopolitan
B. Endemic
C. Migratory
D. Invasive

ANSWER KEY

Correct Option: **B** — Endemic. An endemic species occurs naturally only in a specific region and nowhere else in the world.

Question 94 of 100

Topic: Issues in the Conservation of Biodiversity with Emphasis on Nigerian Species

Q94. Biodiversity conservation is best defined as

A. The exploitation of wildlife for commercial gain
B. The protection, management and sustainable use of living organisms and their habitats
C. The elimination of invasive species only
D. The study of animal fossils

ANSWER KEY

Correct Option: **B** — The protection, management and sustainable use of living organisms and their habitats. Biodiversity conservation involves protecting and sustainably managing organisms and habitats to prevent extinction.

Question 95 of 100

Topic: Issues in the Conservation of Biodiversity with Emphasis on Nigerian Species

Q95. Which of the following is a major cause of biodiversity loss in Nigeria?

A. Habitat destruction and deforestation
B. Increased rainfall
C. Reduced human population
D. Improved wildlife legislation

ANSWER KEY

Correct Option: **A** — Habitat destruction and deforestation. Habitat destruction/deforestation, poaching, pollution and climate change are major causes of biodiversity loss in Nigeria.

Question 96 of 100

Topic: Issues in the Conservation of Biodiversity with Emphasis on Nigerian Species

Q96. Which of the following is an endangered animal species found in Nigeria?

A. Domestic cat
B. Cross River gorilla (Gorilla gorilla diehli)
C. House rat
D. Common housefly

ANSWER KEY

Correct Option: **B** — Cross River gorilla (Gorilla gorilla diehli). The Cross River gorilla is a critically endangered species found in Nigeria's Cross River region.

Question 97 of 100

Topic: Issues in the Conservation of Biodiversity with Emphasis on Nigerian Species

Q97. The protection of species within their natural habitats, such as through national parks, is known as

A. Ex-situ conservation
B. In-situ conservation
C. Captive breeding only
D. Gene banking

ANSWER KEY

Correct Option: **B** — In-situ conservation. In-situ conservation protects species within their natural habitats, e.g. national parks and game reserves.

Question 98 of 100

Topic: Issues in the Conservation of Biodiversity with Emphasis on Nigerian Species

Q98. The protection of species outside their natural habitats, such as in zoos or gene banks, is known as

A. In-situ conservation
B. Ex-situ conservation
C. Habitat restoration only
D. Afforestation

ANSWER KEY

Correct Option: **B** — Ex-situ conservation. Ex-situ conservation protects species outside their natural habitats, e.g. zoos, gene banks and captive breeding.

Question 99 of 100

Topic: Issues in the Conservation of Biodiversity with Emphasis on Nigerian Species

Q99. Which of the following is a national park established in Nigeria for wildlife conservation?

A. Yankari National Park
B. Yellowstone National Park
C. Kruger National Park
D. Serengeti National Park

ANSWER KEY

Correct Option: **A** — Yankari National Park. Yankari National Park (along with Cross River and Kainji Lake National Parks) is located in Nigeria.

Question 100 of 100

Topic: Issues in the Conservation of Biodiversity with Emphasis on Nigerian Species

Q100. CITES is an international agreement whose main role is to

A. Fund national parks only
B. Regulate and monitor international trade in endangered species
C. Set climate change targets
D. Control human population growth

ANSWER KEY

Correct Option: **B** — Regulate and monitor international trade in endangered species. CITES regulates and monitors international trade in wild fauna and flora to prevent threats to their survival.